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Building an Agile Network Infrastructure in the Cloud Using AWS Cloud WAN with Fortive

Learn how the industrial conglomerate Fortive built an efficient, agile wide area network using AWS Cloud WAN.

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70%

Reduced network service delivery costs

35%

Reduction in outages

10x

Increased throughput between network



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Efficient & Agile

Retired on-premises data center hardware



Overview

[Fortive](#) wanted an agile, scalable wide area network (WAN) infrastructure that could centralize networking for its operating companies, which operate in 50 countries across six continents using various technologies. To facilitate connectivity as it merged, acquired, or spun off new companies, Fortive had used multiple technologies within a WAN that operated from large physical data centers. “Specifically with networking, it’s always been a challenge to marry disparate technologies and connection types,” says Nick Mullenix, network leader at Fortive. “We strive to provide excellent service to everyone no matter where their operations.”

Fortive’s business model depends on seamless acquisitions and restructuring of its operating companies. For a network infrastructure that reflected its emphasis on technology and innovation, Fortive turned to Amazon Web Services (AWS). Fortive migrated its entire transit network backbone to AWS using [AWS Cloud WAN](#), which helps organizations build, manage, and monitor global WANs. Using AWS Cloud WAN, Fortive retired its on-premises hardware, simplified and centralized networking, reduced major incidents by 35 percent over 3 months, and cut 70 percent in costs for network service delivery.



Opportunity | Using AWS Cloud WAN for an Agile Network Infrastructure for Fortive

Founded in 2016, Fortive provides intelligent operating software, precision technologies, and advanced healthcare solutions through 18 operating companies across more than 225 global locations. It operated its network from physical data centers that initially cost millions to implement. Over years of operation, the infrastructure continued to accumulate technical debt and increased the risk of interruption to the business. In addition to the technical debt, the infrastructure lacked the agility that Fortive required as it expanded its investments into new geographies and cloud-based offerings. Further, the business was pressuring the IT services organization to develop new services, and streamline time to value of acquisitions. Minimizing the time to only on is critical so that the company can realize the value of its investment within a predetermined time frame. The



technical debt from legacy data centers continued to make automation a challenge for Fortive, leading to loss of business value resulting from extended acquisition integration projects.

Fortive had already been using core AWS services for part of its infrastructure, running compute resources inside [Amazon Virtual Private Cloud](#) (Amazon VPC), which helps organizations define and launch AWS resources in a logically isolated virtual network. Fortive charted a new and innovative course using AWS Cloud WAN and related services, providing a path to retire its legacy data center infrastructure services and migrate its full stack to AWS, including connectivity as a core service offering. For about a year beginning in May 2022, Fortive worked alongside AWS solutions architects and [AWS Professional Services](#), which helps organizations to achieve their desired business outcomes using AWS, to plan and implement the migration of its entire network to the cloud.

Solution | Saving 70% in Network Service Delivery Costs While Reducing Incidents by 35%

To connect each site to AWS Cloud WAN, Fortive uses several AWS Partners who provide physical connections through [AWS Direct Connect](#), which creates a dedicated network connection to AWS. Using AWS Cloud WAN, Fortive performed some limited regional testing and saw immediate improvements. “It almost felt like we teleported the data packets because we didn’t have to account for anything other than getting traffic to the AWS backbone,” says Mullenix. “We broke the dependency that we had on physical infrastructure. Our use of AWS Cloud WAN provided this eureka moment: this is actually going to work and work well.”

Working alongside AWS, Fortive migrated all three of its data centers to AWS, using AWS Cloud WAN to simplify and automate networking operations, including SD-WAN and VPN-based connectivity. “We’ve seen the largest benefit in long-legged connections between network spokes that we need to support our environment,” says Mullenix. Overall, Fortive has experienced about 10 times greater throughput using AWS Cloud WAN as it syncs databases in Asia with file stores located in the western US. In one use case, the company needed to connect sites in Australia and San Diego over its corporate WAN, which at 75 Mbps had been oversubscribed and lacked the capacity to provide adequate communication among systems. By changing its routing technique and using AWS Cloud WAN at a total throughput supported by the infrastructure in Melbourne, Fortive saw a perceived speed increase of 66 times on file operations across this link.

Since closing the last of its network data centers in May 2023, Fortive has cut network service 70 percent while improving reliability and performance. “We’ve saved well over 65 percent in operational expenses,”



says Mullenix. Additionally, the company reduced incident tickets for the network stack by 35 percent. Using AWS Cloud WAN, Fortive experienced 3 months—an entire quarter—without any core network engineering issues.

With the new network, Fortive resolves issues more quickly, cutting mean time to resolution by 50 percent during the first 3 months of implementation. “Because we took some of our network technical debt and complexity out of the environment and used some of the great technology within AWS, the team is about 65 percent as busy as we were before,” says Mullenix.

Outcome | Expanding Reach through a Proactive, Agile Network Infrastructure

Fortive plans to expand to additional AWS Regions, next adding Mumbai through the ap-south-1 AWS Region. “With the knowledge we have gained, implementation becomes a repeatable process,” Mullenix says. “We estimate that 12 months would be a comfortable timeline using an architecture similar to our design.”

Fortive expects that its cloud-based network will simplify the acquisition process as new companies come on board to fuel growth. “For us, AWS Cloud WAN does exactly what a network is designed to do, which is to stay up and running, be simple to maintain, have no technical debt, not be a drag on the rest of the company, and be invisible,” says Mullenix. “Within weeks, we can engineer and deploy a solution that improves the latency, throughput, and performance of a production application. That’s a winner in anybody’s book.”

About Fortive

Fortive is an American industrial technology conglomerate with operating companies in 50 countries, providing intelligent operating software, precision technologies, and advanced healthcare solutions.

AWS Services Used

AWS Cloud WAN

AWS Cloud WAN connects your branch offices, data centers, and Amazon Virtual Private Clouds (Amazon VPCs)—helping you build, manage, and monitor global wide area networks.

[Learn more »](#)

AWS Direct Connect



With AWS Direct Connect, you bypass the public internet and create a private connection to the AWS global network.

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Amazon VPC

Define and launch AWS resources in a logically isolated virtual network using Amazon Virtual Private Cloud (Amazon VPC).

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